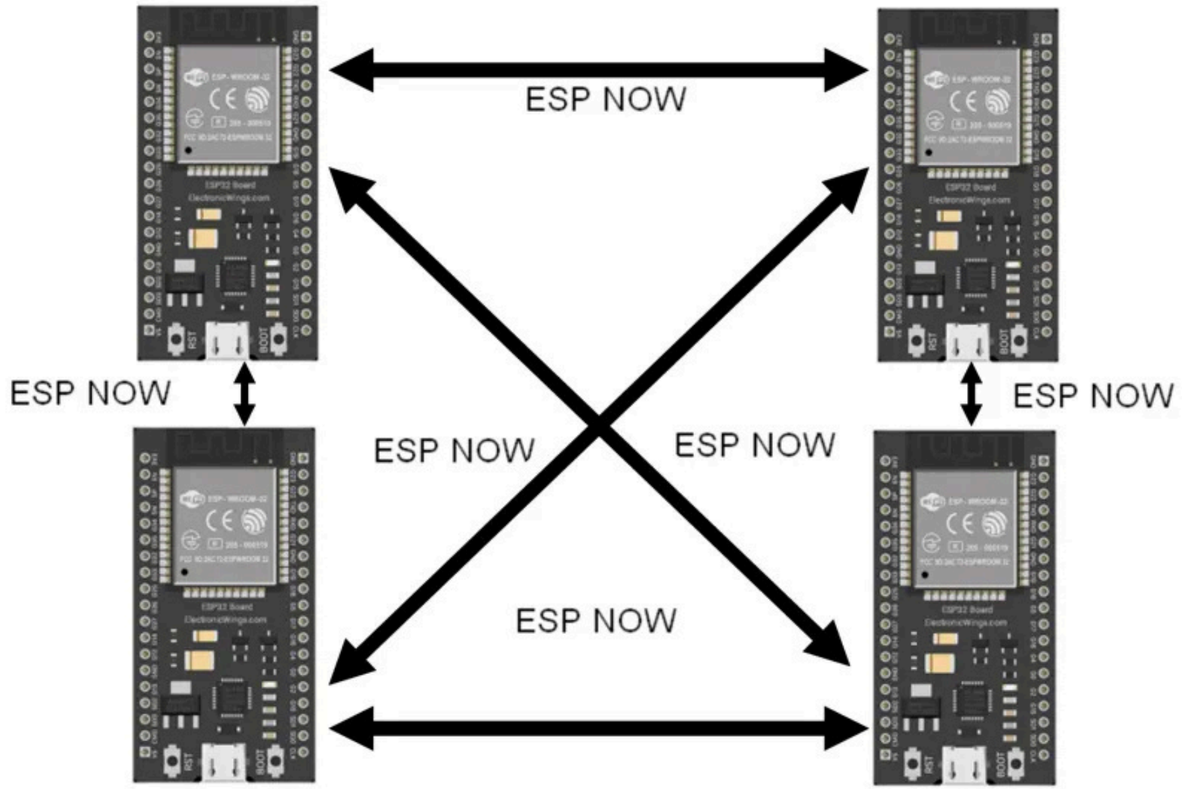
ESP-NOW

Direct, Connectionless Wireless Communication
Between ESP32 Devices

What Is ESP-NOW?

A connectionless wireless protocol from Espressif that lets ESP devices talk directly to each other — riding on the Wi-Fi radio, without joining a Wi-Fi network or router.



Rides the Wi-Fi Radio

Uses the same 2.4GHz radio as Wi-Fi, but data travels in vendor-specific action frames, not normal Wi-Fi traffic.

No Access Point Needed

Devices communicate peer-to-peer — there's no router or Wi-Fi network to join.

Built for Low Latency

Designed for quick, local bursts of data rather than sustained high-throughput links.

Flexible Topology

Supports one-to-one, one-to-many, and many-to-many arrangements — a device can send and receive at once.

Where Is ESP-NOW Used?

Anywhere nearby devices need to trade small bits of data, fast and without infrastructure.

Sensor Networks . Remote Controls . Low-Power IoT . Local Device Control . Mesh-Style Links . Smart Home Devices

Prerequisites & Hardware :

Hardware: One or more ESP32 development boards. For the basic two-device test, use two ESP32 boards.

USB data cable: Required for programming and Serial Monitor access. Make sure the cable supports data, not only charging.

Power: USB power is sufficient for normal testing. A separate regulated supply can be used if the board's power-input requirements are satisfied.

Software: Arduino IDE installed on your computer

Step 1: Configure the Arduino IDE

Step 2: Find the MAC Address of Each ESP32

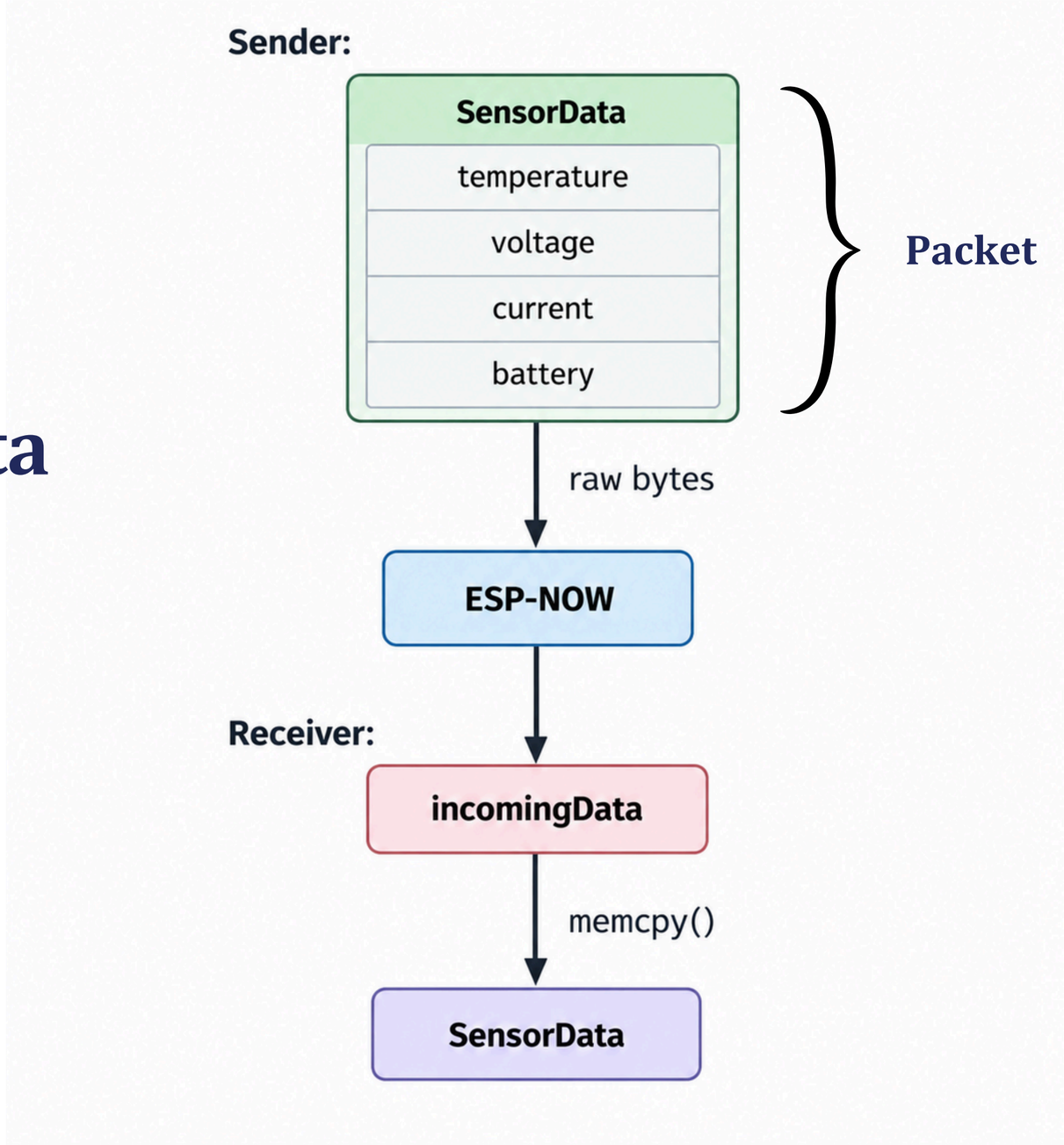
Step 3: Configure the ESP-NOW Receiver

Step 4: Configure the ESP-NOW Sender

Step 5: Run the Trial and Verify

Step 6: Send an Information / Data Packet

Understand the Data Packet



Broadcast Communication

Broadcast sends a packet to all ESP-NOW devices listening on the appropriate channel rather than to one specific MAC address.

Unicast :

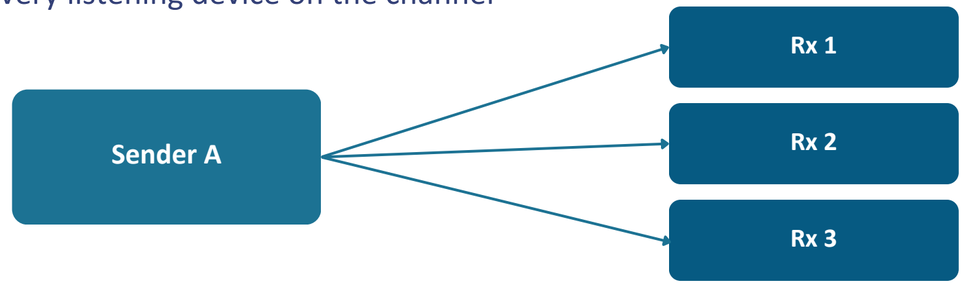
One specific device, addressed directly



Destination = B's exact MAC address

Broadcast :

Every listening device on the channel



Destination = FF:FF:FF:FF:FF:FF

A specific MAC address is unicast. FF:FF:FF:FF:FF:FF is the broadcast destination. Do not name a variable broadcastAddress and then put a specific peer MAC inside it; the destination value determines the actual addressing mode

That's ESP-NOW

The access the implementation guide
Scan the QR / click on the link in the description

Thank You